Phase 154: Coaching Proton Flow Audit

Security Audit & Bug Fix Verification Report

17/17 TESTS PASSED

HC Portal — KPB Pertamina

Tanggal: 11 Maret 2026

Auditor: Claude Code (Automated)

1. Ringkasan Eksekutif

Hasil: LULUS — Semua 17 Test Case Passed

Phase 154 melakukan audit keamanan dan perbaikan bug pada modul Coaching Proton. Audit mencakup 7 area requirement (PROTON-01 s/d PROTON-07). Ditemukan **3 security fix** dan **1 bug fix** yang langsung diperbaiki. Semua perbaikan telah diverifikasi melalui browser testing.

Metrik	Nilai
Total Test Cases	**17**
Passed	**17**
Failed	0
Issues Found & Fixed	**4**
Requirements Covered	PROTON-01 ~ PROTON-07
Files Modified	CDPController.cs, AdminController.cs

2. Perbaikan yang Diterapkan

FIX-1: UploadEvidence IDOR (Security - Critical)
Coach mana pun bisa POST ke `/CDP/UploadEvidence?progressId=X` untuk coachee yang bukan miliknya. Diperbaiki dengan menambahkan validasi `CoachCoacheeMappings`.

FIX-2: DownloadEvidence Coach Scope (Security)
Akses download evidence menggunakan section-equality check, bukan mapping check. Diganti dengan `CoachCoacheeMappings.AnyAsync()` untuk konsistensi.

FIX-3: HCReviewFromProgress Idempotency (Edge Case)
AJAX endpoint HC review tidak memiliki guard duplikat yang ada di versi non-AJAX.

Ditambahkan guard `HCApprovalStatus != "Pending"`.

FIX-4: SubmitInterviewResults Missing ProtonFinalAssessment (Bug)
Tahun 3 interview yang lulus tidak membuat record `ProtonFinalAssessment`, sehingga status "Lulus" tidak muncul di Histori Proton dan dashboard. Diperbaiki dengan menambahkan insert `ProtonFinalAssessment` setelah interview selesai.

3. Detail Hasil Testing

3.1 Coach-Coachee Mapping (PROTON-01)

#	Test Case	Hasil
1	Create Coach-Coachee Mapping	PASS
2	Coachee Sees Deliverables Immediately	PASS
3	Mapping Reactivation Restores Deliverables	PASS
4	Multi-Unit User Mapping	PASS

3.2 Evidence Upload (PROTON-02)

#	Test Case	Hasil
5	Coach Uploads Evidence for Coachee	PASS
6	Coach Cannot Upload Evidence for Unmapped Coachee (IDOR Fix)	PASS

3.3 Coaching Sessions (PROTON-03)

#	Test Case	Hasil
7	Coach Creates Coaching Session	PASS

3.4 Approval Chain (PROTON-04)

#	Test Case	Hasil
8	SrSpv Sees Only Scoped Deliverables	PASS
9	SectionHead Rejects Deliverable with Reason	PASS
10	Worker Cannot Access Approval Endpoints	PASS

3.5 HC Oversight (PROTON-05)

#	Test Case	Hasil

11	HC Reviews All Coachees (cross-section)	**PASS**

3.6 Assessment Proton (PROTON-06)

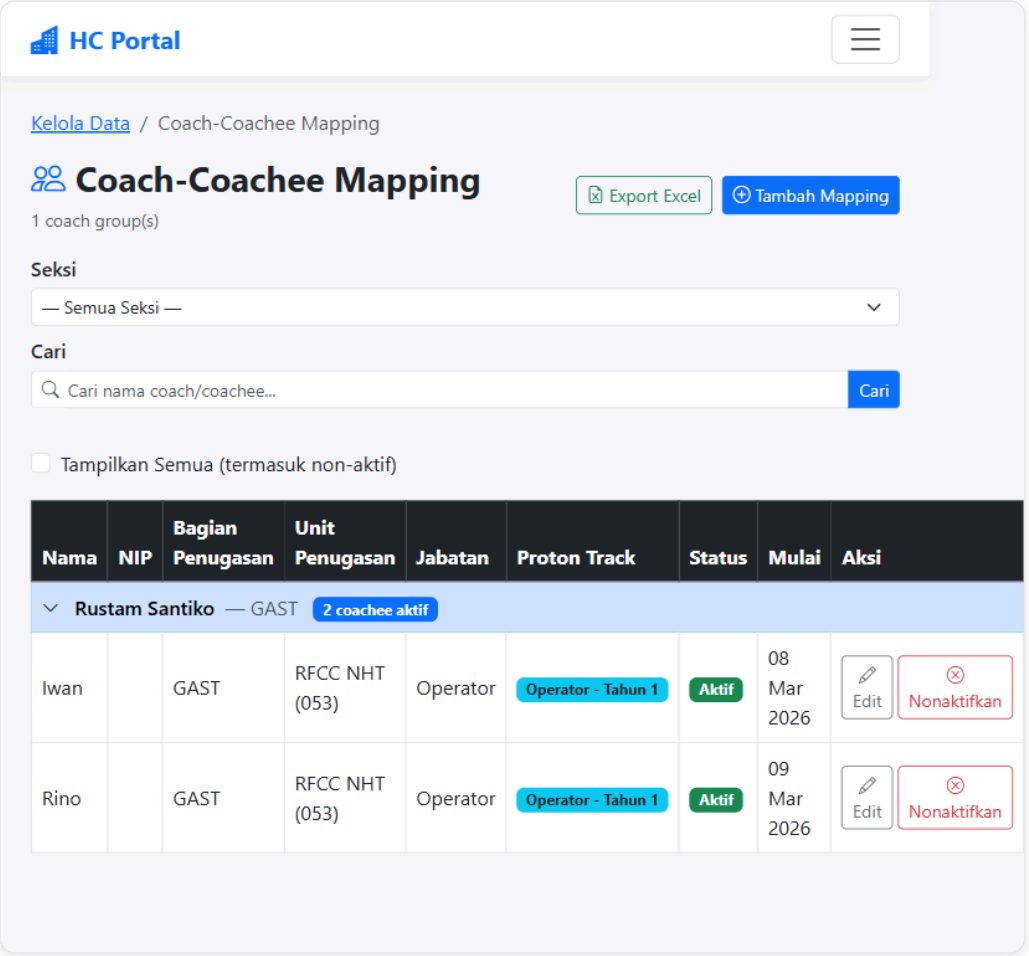
#	Test Case	Hasil
12	Create Assessment Proton — Tahun 1-2 Online	**PASS**
13	Create Assessment Proton — Tahun 3 Interview	**PASS**
14	Submit Interview Results Creates ProtonFinalAssessment (Bug Fix)	**PASS**

3.7 Histori Proton (PROTON-07)

#	Test Case	Hasil
15	Histori Proton Timeline Completeness	**PASS**
16	Histori Proton Authorization (IDOR blocked)	**PASS**
17	Mapping Endpoint Authorization	**PASS**

4. Bukti Screenshot

4.1 Coach-Coachee Mapping (HC View)



Gambar 1: Halaman mapping coach-coachee menampilkan 2 coachee aktif di bawah Coach Rustam

4.2 Coaching Proton — HC Oversight View

HC Portal

Coaching Proton

Monitor deliverable progress and approval status

← Kembali

Semua Bagian | Semua Unit

— Pilih Coachee — | Semua Track | Semua Tahun

Cari kompetensi... | Reset

Menampilkan 1–4 dari 4 deliverable

Progress
50%

Pending Actions
2

Pending Approvals
0

Coachee	Kompetensi	Sub Kompetensi	Deliverable	Evidence
Iwan	1.1 Safe Work Practice	4. Mampu memahami 9 Perilaku Wajib.	4. Menjelaskan 9 Perilaku Wajib.	Sudah Upload
		5. Mampu memahami Grafik Flammable Range, sumber dan jenis-jenis Rambatan Panas.	5. Menjelaskan : a. Suhu penyalaan dan Titik Nyala b. Batasan Mudah Terbakar pada suatu gas (LEL/LFL & UEL/UFL) c. Sumber dan Jenis Rambatan Panas	Belum Upload
		4. Mampu memahami 9 Perilaku Wajib.	4. Menjelaskan 9 Perilaku Wajib.	Belum Upload
			5. Menjelaskan	Belum Upload

Rino	**1.1 Safe Work Practice**	5. Mampu memahami Grafik Flammable Range, sumber dan jenis-jenis Rambatan Panas.	: a. Suhu penyalaan dan Titik Nyala b. Batasan Mudah Terbakar pada suatu gas (LEL/LFL & UEL/UFL) c. Sumber dan Jenis Rambatan Panas	

Antrian Review HC **1 pending**

Coachee	Kompetensi	Sub-Kompetensi	Deliverable	Status	Tangg Subm
Iwan	1.1 Safe Work Practice	4. Mampu memahami 9 Perilaku Wajib.	4. Menjelaskan 9 Perilaku Wajib.	Approved	09 Ma 2026

Setelah diklik Review, deliverable ditandai sebagai sudah diperiksa HC. Status tidak berubah.

Gambar 2: HC melihat semua deliverable dari semua coachee (cross-section), dengan progress 50% dan antrian review

4.3 Assessment Proton — Tahun 3 Interview Result

HC Portal

Kelola Data / Assessment Monitoring / UAT Proton Tahun 3 Interview

← Back to Monitoring

Assessment Proton Closed

UAT Proton Tahun 3 Interview

20 Mar 2026

1	1	0	0
Total Assigned	Completed	In Progress	Not Started

Per-User Status

Export Results | Force Close All

Name	Progress	Status	Score	Result	Completed At	T R
Iwan	—	Completed	—	Pass	11 Mar 2026 11.28	–

Last updated: 19.33.37

Input Hasil Interview Proton — Tahun 3

Iwan Lulus

Daftar Juri

Dr. Ahmad, Ir. Budi

Penilaian per Aspek (1=Kurang, 5=Luar Biasa)

Pengetahuan Teknis

4 — Sangat Baik

Kemampuan Operasional

4 — Sangat Baik

Keselamatan Kerja

5 — Luar Biasa

Komunikasi & Kerjasama

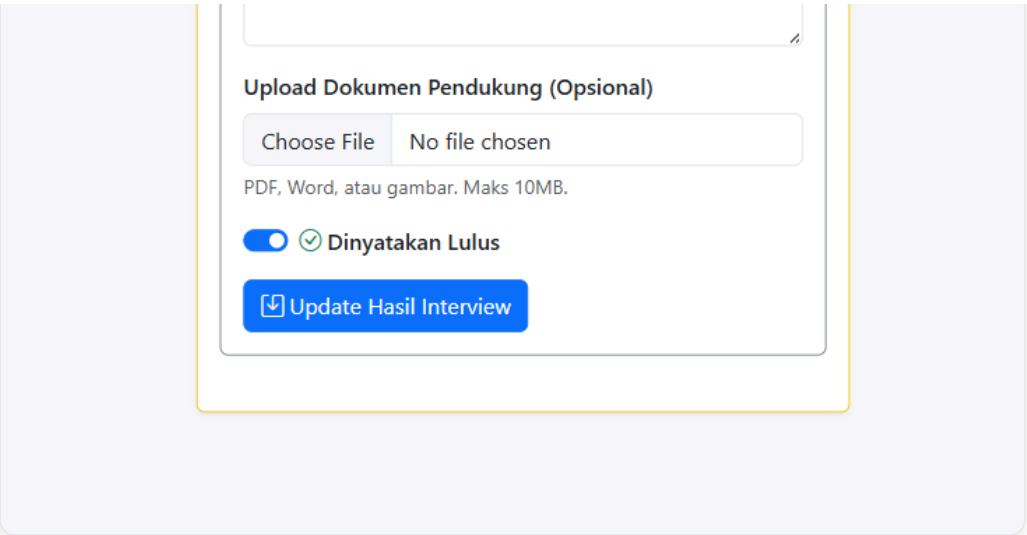
4 — Sangat Baik

Sikap Profesional

4 — Sangat Baik

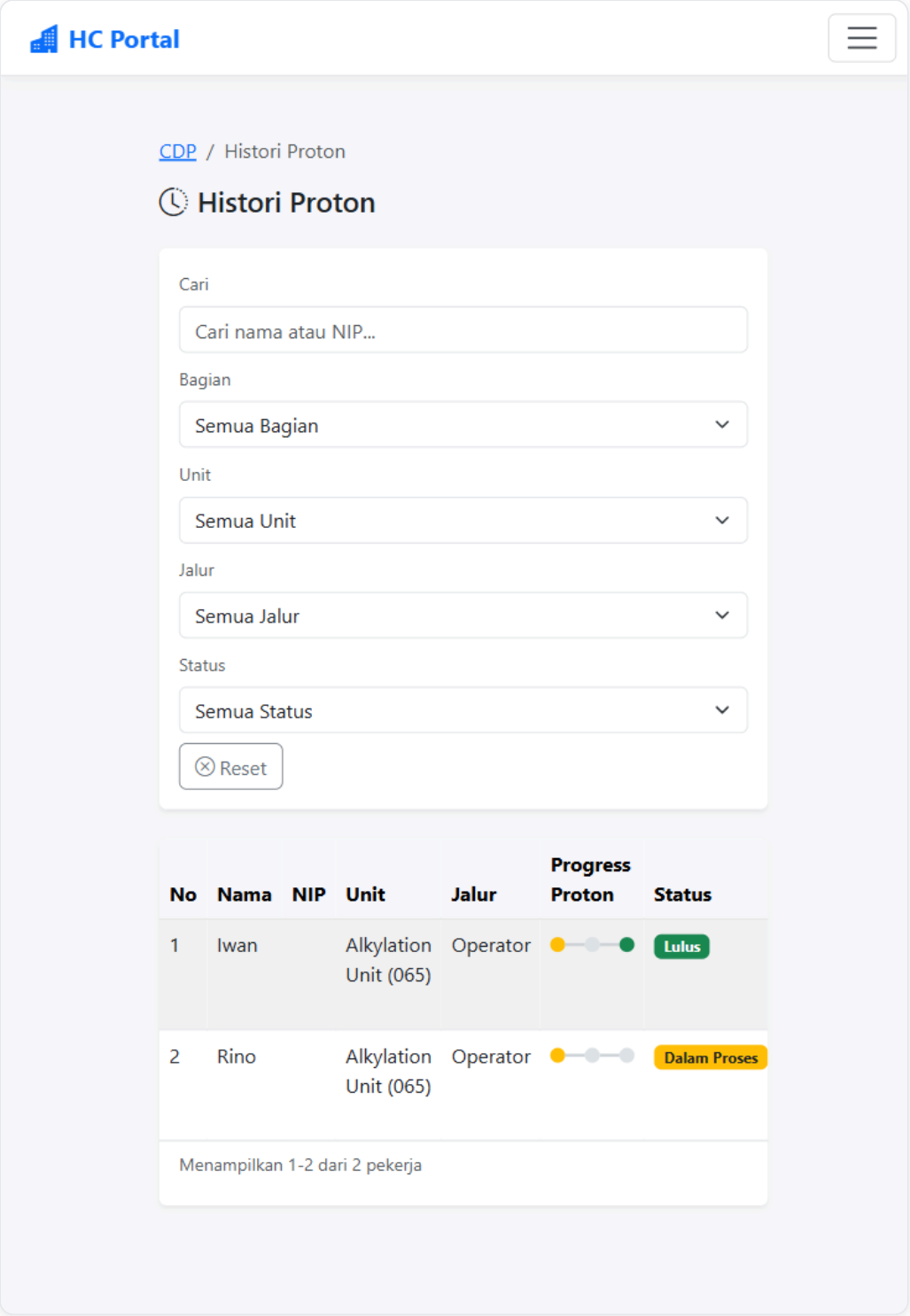
Catatan

Kandidat menunjukkan penguasaan teknis yang baik



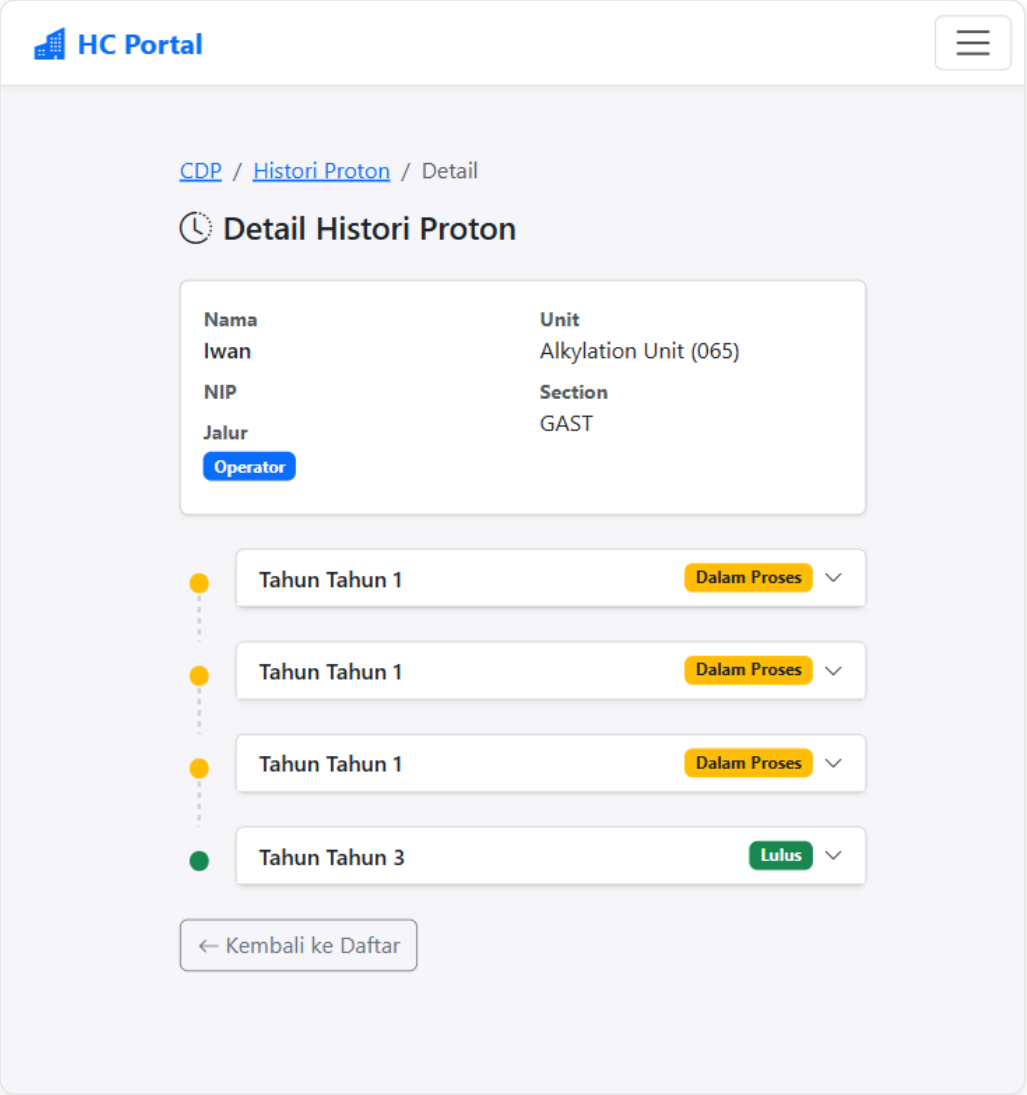
Gambar 3: Hasil interview Tahun 3 berhasil disimpan — Iwan status Completed/Pass dengan penilaian per aspek

4.4 Histori Proton — Daftar Coachee



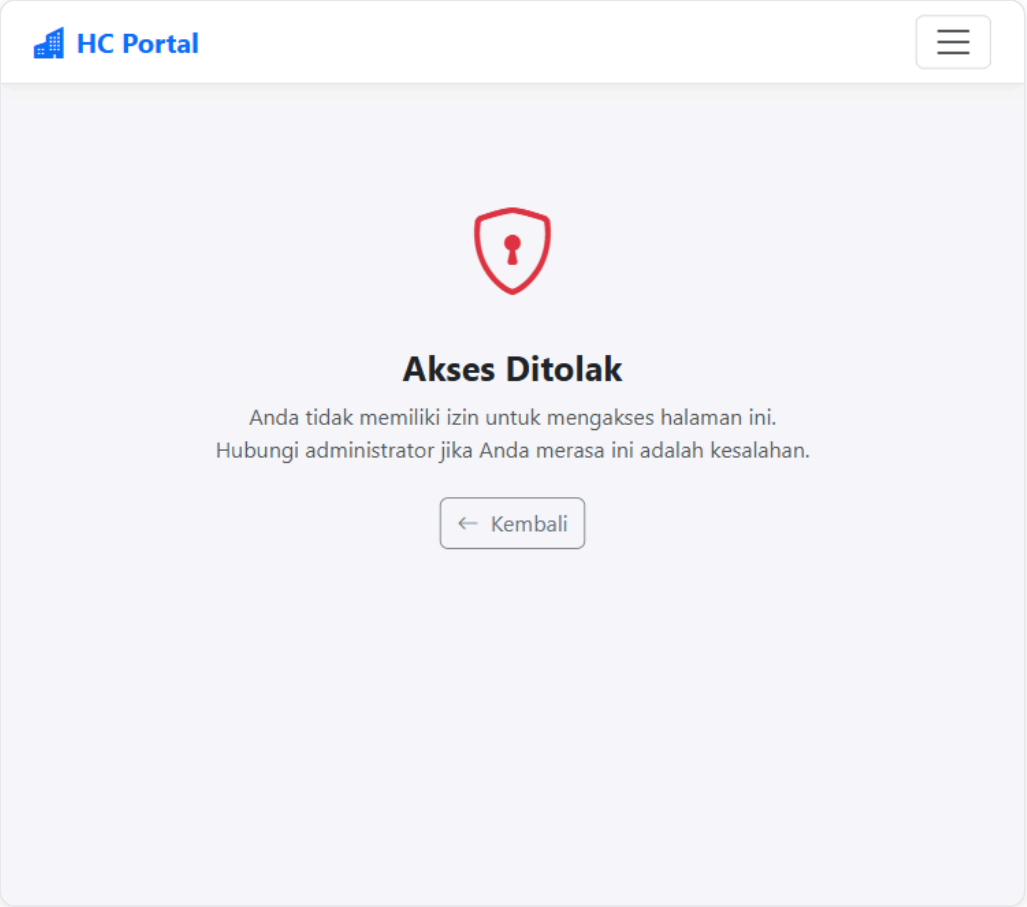
Gambar 4: Iwan menunjukkan status "Lulus" setelah ProtonFinalAssessment dibuat (bug fix verified)

4.5 Histori Proton — Timeline Detail



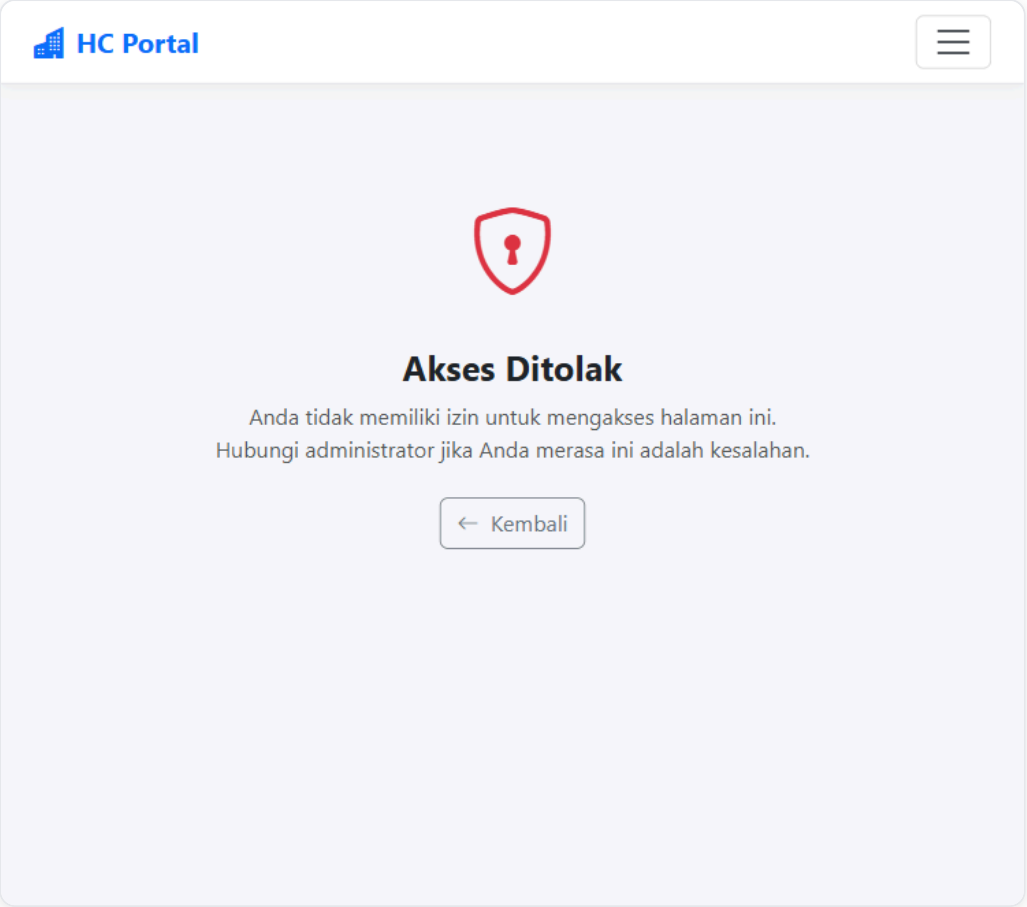
Gambar 5: Timeline Iwan menampilkan 3x Tahun 1 (Dalam Proses) dan Tahun 3 (Lulus)

4.6 Authorization — IDOR Protection (Histori Proton)



Gambar 6: Coachee Iwan mencoba akses HistoriProtonDetail user lain — "Akses Ditolak"

4.7 Authorization — Admin Endpoint Protection



Gambar 7: Coachee Iwan mencoba akses Admin/CoachCoacheeMapping — "Akses Ditolak"

5. Kesimpulan

Phase 154 — VERIFIED & COMPLETE

- **Security:** 3 vulnerability diperbaiki (IDOR upload evidence, download scope, idempotency guard)
- **Bug Fix:** 1 bug kritis diperbaiki (ProtonFinalAssessment tidak dibuat saat interview lulus)
- **Authorization:** Semua endpoint terproteksi — coachee tidak bisa mengakses data user lain atau halaman admin
- **Functional:** Semua flow berjalan normal — mapping, evidence upload, coaching session, approval chain, assessment, dan histori

Commits

Hash	Message
`02897c0`	Security fixes: UploadEvidence IDOR, DownloadEvidence scope, HCReviewFromProgress idempotency
`e95a36b`	Bug fix: SubmitInterviewResults now creates ProtonFinalAssessment on pass
`16c10ac`	Fix: GetEligibleCoachees returns all assigned coachees for Tahun 3 tracks
`cf9c1ba`	UAT complete: 17/17 passed, 0 issues

— End of Report —

Generated: 11 Maret 2026 | HC Portal KPB